

(1)

$$0 \leq t \leq t_a$$

$$\dot{q}(t) = c_0 + c_1 t$$

$$\dot{q}(t) = 0 + at \quad \{ \dot{q}(0) = 0, \ddot{q}(t) = a \}$$

$$\text{Also } \dot{q}(t_a) = at_a = v \quad \Rightarrow a = \frac{v}{t_a}$$

$$q(t) = \bar{c}_0 + \frac{vt^2}{2t_a}$$

$$q(t) = q_0 + \frac{vt^2}{2t_a} \quad \{ q(0) = q_0 \} \quad \textcircled{I}$$

$$q(t_a) = q_0 + \frac{vt_a}{2} \quad \text{--- ①}$$

$$t_a \leq t \leq t_f - t_a$$

$$\dot{q}(t) = v$$

$$q(t) = \bar{C}_1 + vt$$

$$q(t_{f/2}) = \bar{C}_1 + \frac{vt_f}{2} = \frac{q_0 + q_f}{2}$$

$$\bar{C}_1 = \frac{q_0 + q_f - vt_f}{2}$$

$$q(t) = \frac{q_0 + q_f - vt_f}{2} + vt \quad - (2) \quad \textcircled{\text{II}}$$

From ① and ②

$$q(t_a) = \frac{q_0 + q_f - vt_f}{2} + vt_a = q_0 + \frac{vt_a}{2}$$

Solve for t_a

$$t_a = \frac{(q_0 - q_f + vt_f)}{v} \quad - \textcircled{*}$$

$$t_f - t_a \leq t \leq t_f$$

$$\dot{q}(t) = c_2 + c_3 t$$

$$\dot{q}(t) = a(t_f - t) \quad \{ \ddot{q}(t) = -a ; \dot{q}(t_f) = 0 \}$$

$$\dot{q}(t) = \frac{v}{t_a} (t_f - t) \quad \{ \text{As } a = v/t_a \}$$

$$q(t) = \bar{c}_2 - \frac{1}{2} \frac{v}{t_a} (t_f - t)^2$$

$$q(t) = q_f - \frac{v}{2t_a} (t_f - t)^2 \quad \{ q(t_f) = q_f \} \quad \textcircled{\pi}$$

(a) From (I), (II), (III)

$$q(t) = \begin{cases} q_0 + \frac{vt^2}{2t_a} & 0 \leq t < t_a \\ \frac{q_0 + q_f - vt_f}{2} + vt & t_a \leq t < t_f - t_a \\ q_f - \frac{v}{2t_a} (t_f - t)^2 & t_f - t_a \leq t < t_f \end{cases}$$

where $t_a = \frac{(q_0 - q_f + vt_f)}{v}$ From (*)

(b) For the constant V to be possible

$$0 < t_a \leq \frac{t_f}{2}$$

$$0 < \frac{q_0 - q_f + V t_f}{V} \leq \frac{t_f}{2}$$

From (*)

$$0 < \frac{q_0 - q_f}{V} + t_f \leq \frac{t_f}{2}$$

$$-t_f < \frac{q_0 - q_f}{V} \leq -\frac{t_f}{2}$$

$$\frac{t_f}{2} \leq \frac{q_f - q_0}{V} < t_f$$

$$\frac{(q_f - q_0)}{t_f} < V \leq \frac{2(q_f - q_0)}{t_f}$$